VS School

Student Management Portal

Manage Students, Teachers, and Resources Efficiently with Our Comprehensive School Portal

Introduction & Project Overview

- Web-based student management system for a school (VS School).
- Provides separate experiences for Students and Admin/Teachers.
- Students can register and view their details.
- Admins can search, filter, update, delete, and export student records.

Objectives

- Developed a mini-project using HTML, java Eclipse and PostgreSQL.
- Implement complete student lifecycle – registration, view, update, delete.
- Demonstrate core OOP concepts (Encapsulation, Inheritance, Polymorphism, Abstraction).
- Provide a clean, easy-to-use interface for both students and administrators.

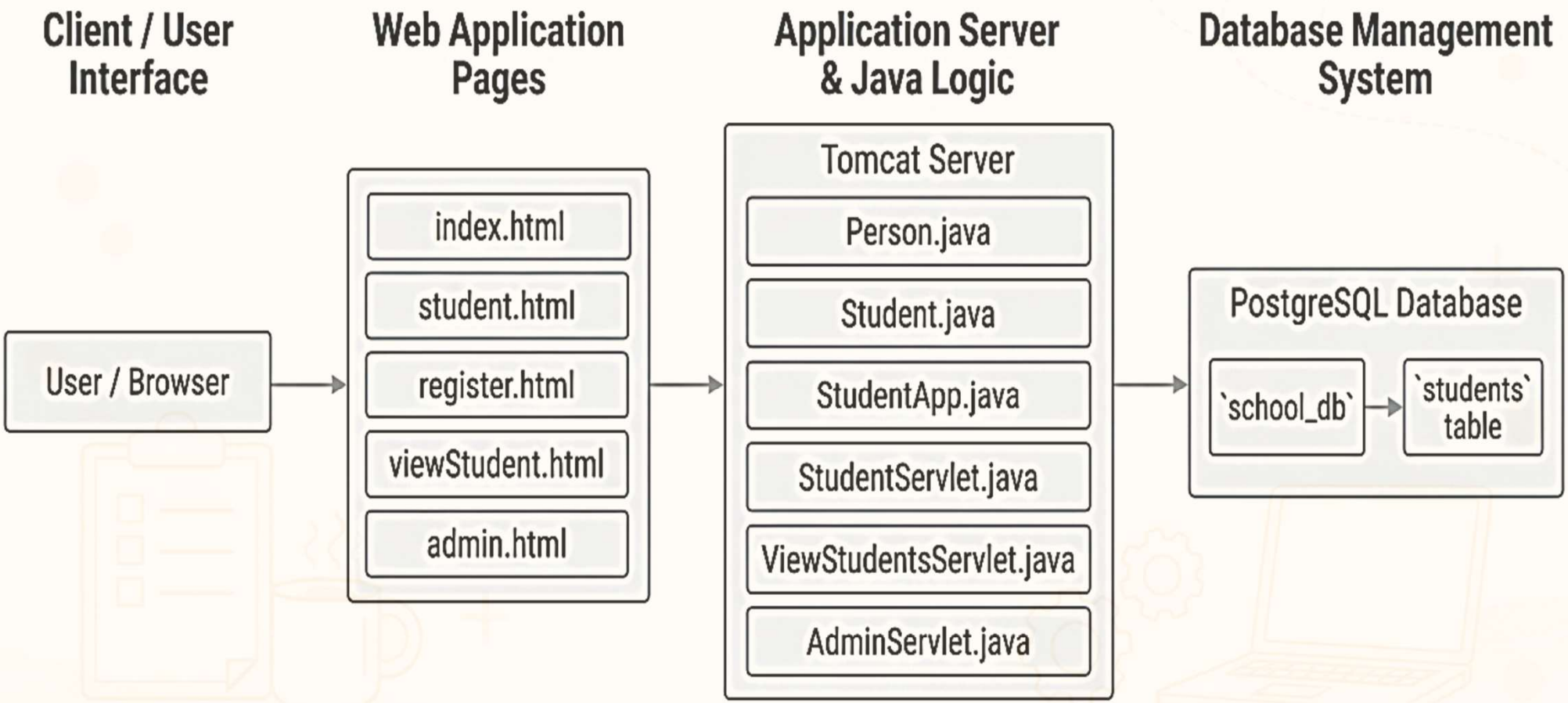
Technologies Used

- Frontend: HTML5, CSS3, JavaScript
- Backend: Java
- Database: PostgreSQL
- API: PostgreSQL JDBC Driver (Version: 42.7.10)
- Server: Apache Tomcat v9.0
- Tools: Eclipse IDE (Version: 2025-12 (4.38.0)), pgAdmin 4 (Version: 9.12)
- GitHub: https://github.com/vaibhavshinde144/Student-Management-Portal.git

System Architecture – Flow

1. Browser Link (index.html) - VS School - Student Management Portal.
2. Student Portal Button (student.html)>
   - Enter Student Id and View Student Details
   - Register New Student (register.html) > Student fills the registration form; data is submitted to StudentServlet for validation and insertion into DB
3. Admin/Teachers Portal Button >
   - Admin enters username and password to access the Admin Portal (login.html)
   - Admin (admin.html) can filter students, update and delete existing students, add new student and export data using AdminServlet. Updated into DB

Project Structure



HTML Files Overview

- index.html – Home page with banner and navigation buttons.
- student.html – Student Portal: enter Student ID, view details, link to registration, Home button.
- register.html – Student Registration Form with client-side validation, register button, home button, reset Button
- login.html – Admin Login screen to access Admin Portal.
- admin.html – Admin dashboard with filters, table, pagination, update/delete button, export excel file, Add student button, home button.
- viewStudent.html – Simple view page for a single student (via servlet output).

Java Files Overview

- Person.java – Abstract base class storing personId, firstName, lastName.
- Student.java – Extends Person; adds email, phone, dob, gender, address, course, age and display() method.
- StudentApp.java – Console main class using ArrayList<Person> to demonstrate polymorphism.
- StudentServlet.java – Handles registration, validation, duplicate checks, and insert into students table.
- AdminServlet.java – Handles filtering, pagination, update, delete, and CSV export.
- ViewStudentsServlet.java – Displays details of a single student based on student_id.

pgAdmin & Database Overview

- Database: school_db (PostgreSQL) managed using pgAdmin 4.
- Core table: students

```
student_id    SERIAL        PRIMARY KEY,
first_name    VARCHAR(20)   CHECK ([A-Za-z]+$),
last_name     VARCHAR(30)   CHECK ([A-Za-z]+$),
email         VARCHAR(50)   UNIQUE,
phone         VARCHAR(10)   UNIQUE                    CHECK (^[0-9]{10}$),
dob           DATE          CHECK (dob < CURRENT_DATE),
gender        VARCHAR(10),
address       VARCHAR(200),
course        VARCHAR(50),
age           INTEGER,
```

- Add and view operations from student portal and all CRUD operations from admin/teacher portal .

Conclusion

- The VS School Student Management Portal demonstrates a complete Java web mini-project.
- It integrates HTML/JavaScript front-end pages with Java Servlets and a PostgreSQL database.
- The design uses OOP principles, server-side validation, pagination, and CSV export.
- This project can be extended to include teacher modules, authentication from database, and additional reports.